Program to check Strong Number

Last Updated : 23 Jul, 2025

Strong Numbers are the numbers whose sum of factorial of digits is equal to the original number. Given a number, check if it is a Strong Number or not.

Examples:

```
Input  : n = 145
Output : Yes
Sum of digit factorials = 1! + 4! + 5!
                        = 1 + 24 + 120
                        = 145

Input :  n = 534
Output : No
```

```
1) Initialize sum of factorials as 0.
2) For every digit d, do following
   a) Add d! to sum of factorials.
3) If sum factorials is same as given
   number, return true.
4) Else return false.
```

An optimization is to precompute factorials of all numbers from 0 to 10.

C++ Java Python3 C# JavaScript PHP

```cpp
// C++ program to check if a number is
// strong or not.
#include <bits/stdc++.h>
using namespace std;

int f[10];

// Fills factorials of digits from 0 to 9.
void preCompute()
{
    f[0] = f[1] = 1;
```

```
    for (int i = 2; i<10; ++i)
        f[i] = f[i-1] * i;
}

// Returns true if x is Strong
bool isStrong(int x)
{
    int factSum = 0;

    // Traverse through all digits of x.
    int temp = x;
    while (temp)
    {
        factSum += f[temp%10];
        temp /= 10;
    }

    return (factSum == x);
}

// Driver code
int main()
{
    preCompute();

    int x = 145;
    isStrong(x) ? cout << "Yes\n" : cout << "No\n";
    x = 534;
    isStrong(x) ? cout << "Yes\n" : cout << "No\n";
    return 0;
}
```

Output

```
Yes
No
```

Time Complexity: O(logn)

Auxiliary Space: O(1), since constant extra space has been taken.

Approach#2: Using Iterative Method

This approach converts the input number into a list of its digits. It then computes the factorial sum of each digit and adds them up. If the final sum is equal to the input number, it returns "Yes", otherwise it returns "No".

Algorithm

1. Define a function is_strong(n) that takes a number n as input.
2. Convert the number to a string and get its digits.
3. Calculate the factorial of each digit using an iterative method.
4. Sum the factorials of all digits.
5. If the sum is equal to the number n, return "Yes", otherwise return "No".

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```
// C++ code addition

#include <iostream>
#include <vector>
#include <string>
using namespace std;

string is_strong(int n) {
  // Convert the number to an array of digits
    vector<int> digits;
    int temp = n;
  // Loop through each digit
    while (temp != 0) {
        digits.insert(digits.begin(), temp % 10);
        temp /= 10;
    }
    int factorial_sum = 0;
  // Calculate the factorial of the digit
    for (int d : digits) {
        int f = 1;
        for (int i = 1; i <= d; i++) {
            f *= i;
        }
      // Add the factorial to the sum
        factorial_sum += f;
    }
```

```
    // Check if the sum of factorials is equal to the original
number
    if (factorial_sum == n) {
        return "Yes";
    } else {
        return "No";
    }
}

int main() {
    int n = 145;
    cout << is_strong(n) << endl;
    return 0;
}

// The code is contributed by Arushi Goel.
```

Output

```
Yes
```

Time Complexity: O(kn^2), where k is the number of digits in the number and n is the maximum value of a digit

Auxiliary Space: O(k), where k is the number of digits in the number.

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